

A PROJECT TITLE
ON
Wine Quality Prediction Using Ridge, Lasso, and Elastic Net Regression
project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF _____

Name of Title:

1. Abstract:

should be 300 words

Machine Learning techniques have become increasingly important in predicting product quality using data-driven approaches. This project focuses on predicting the quality of wine by analyzing its physicochemical properties such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, sulphates, density, pH, and alcohol content. The objective is to build an accurate regression model that estimates wine quality while minimizing prediction error.

The dataset is first preprocessed by handling missing values, performing feature scaling using StandardScaler, and separating training and testing data. Three regularized linear regression models, namely Ridge Regression (L2 Regularization), Lasso Regression (L1 Regularization), and Elastic Net Regression (Combination of L1 and L2), are implemented to reduce overfitting and improve model generalization. The models are trained using appropriate regularization parameters and their performance is compared.

The effectiveness of each model is evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The project also analyzes the impact of regularization on feature selection, coefficient shrinkage, and prediction accuracy. By comparing Ridge, Lasso, and Elastic Net, the proposed system identifies the most suitable regression technique for predicting wine quality. This project demonstrates the practical application of regularized machine learning models for reliable quality prediction and supports data-driven decision making in the food and beverage industry.

2. INTRODUCTION:

Wine quality prediction is an important application of Machine Learning that helps estimate the quality of wine using its physicochemical properties such as acidity, pH, density, sulphates, and alcohol content. Instead of relying only on manual testing, machine learning models can analyze historical data and predict wine quality more efficiently and accurately.

In this project, Ridge Regression, Lasso Regression, and Elastic Net Regression are used to build and compare prediction models. The dataset is preprocessed using feature scaling and then evaluated using performance metrics such as MSE, RMSE, and R^2 Score to identify the model that provides the best prediction accuracy.

3. SOFTWARE REQUIREMENTS:

The proposed Wine Quality Prediction system is developed using Python 3.x as the programming language with Jupyter Notebook, Visual Studio Code, or Google Colab as the development environment. The project uses the Wine Quality Dataset (CSV) for training and testing the machine learning models. Python libraries such as NumPy and Pandas are used for data preprocessing, Scikit-learn is used to implement the Ridge, Lasso, and Elastic Net Regression algorithms, and Matplotlib and Seaborn are used for data visualization and performance analysis. The system can be executed on a computer running Windows 10/11 with a minimum of 4 GB RAM, an Intel Core i3 or higher processor, and at least 500 MB of free storage space.

Signature of the Guide

Course Instructor